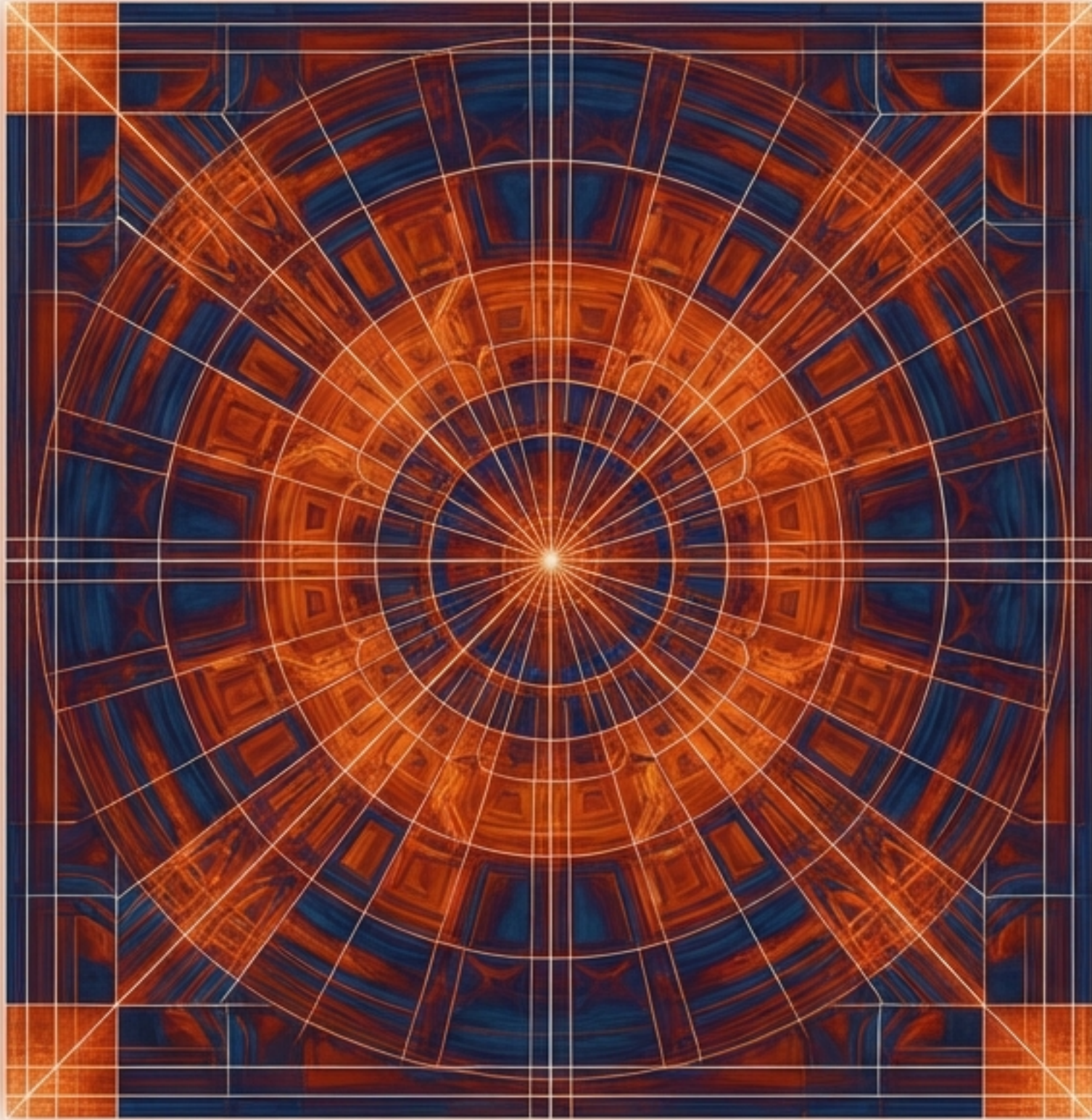


When Visual AI Meets Market Reality

A 296K-Parameter CNN vs. The Coin Flip



VTS Tournament Submission:
Stacking Sats Challenge

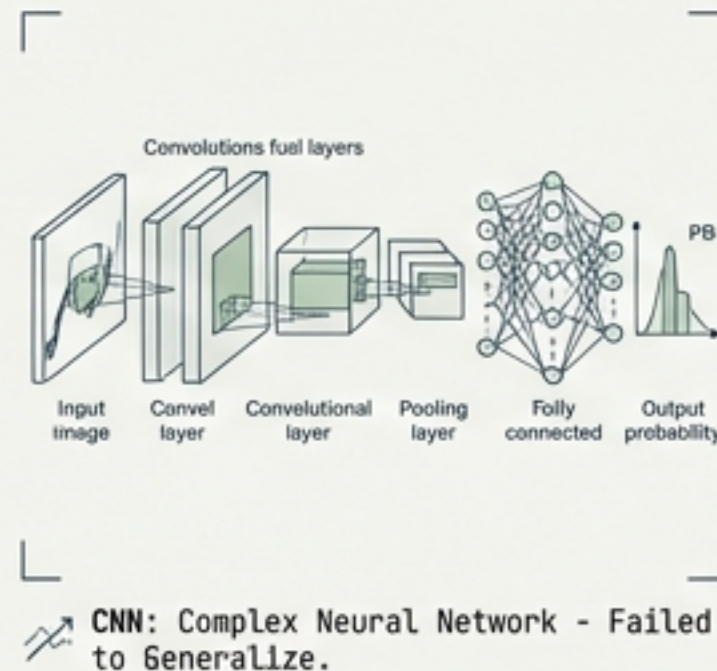
EXECUTIVE SUMMARY: SCIENTIFIC INTEGRITY & ALGORITHMIC SIMPLIFICATION

THE CONTEXT

THE BOTTOM LINE: SCIENTIFIC INTEGRITY > RAW PERFORMANCE

We engineered a technically sophisticated system to predict Bitcoin allocation using visual deep learning (CNNs). However, rigorous post-mortem ablation studies revealed that the model performed equivalently to a random process.

Accuracy (Test): 50.1% (Random Baseline).



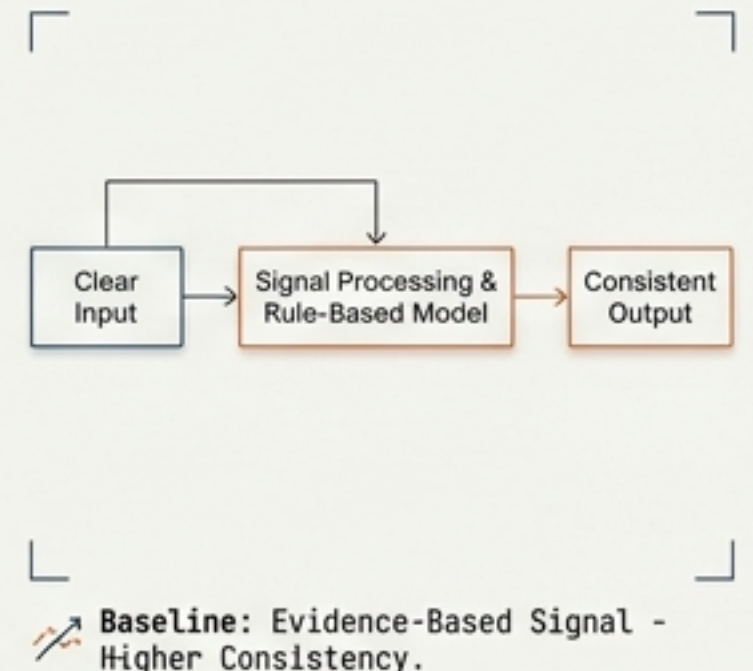
THE RESOLUTION

THE PIVOT

Instead of forcing a complex failure, we pivoted to a “Simplified Baseline”—an evidence-based approach that stripped away the AI to achieve higher consistency.

This project serves as a case study in why signal quality matters more than model complexity in financial time series.

Consistency (Test): >85% (Simplified Baseline).



“The best model is the one that works, not the one that sounds impressive.”

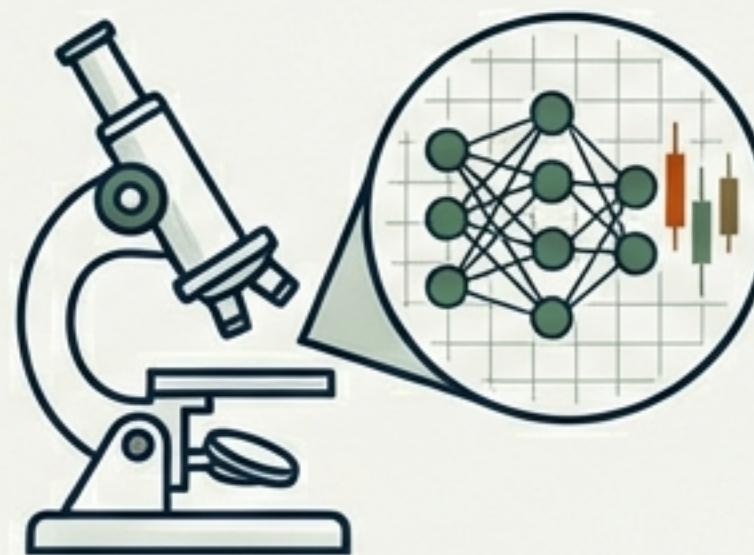
The Hypothesis: Automating the Trader's Eye



The Foundation (MBA Era)

Inspired by Magee & Edwards' Technical Analysis.
Human traders 'see' patterns that quant models miss.

JetBrains Mono
Visual Intuition.



The Discovery (OMSA Program)

JP Morgan & GT ML4T
Research: CNNs achieving
0.90+ F1 scores on candlestick
pattern classification.

JetBrains Mono
Academic Success.

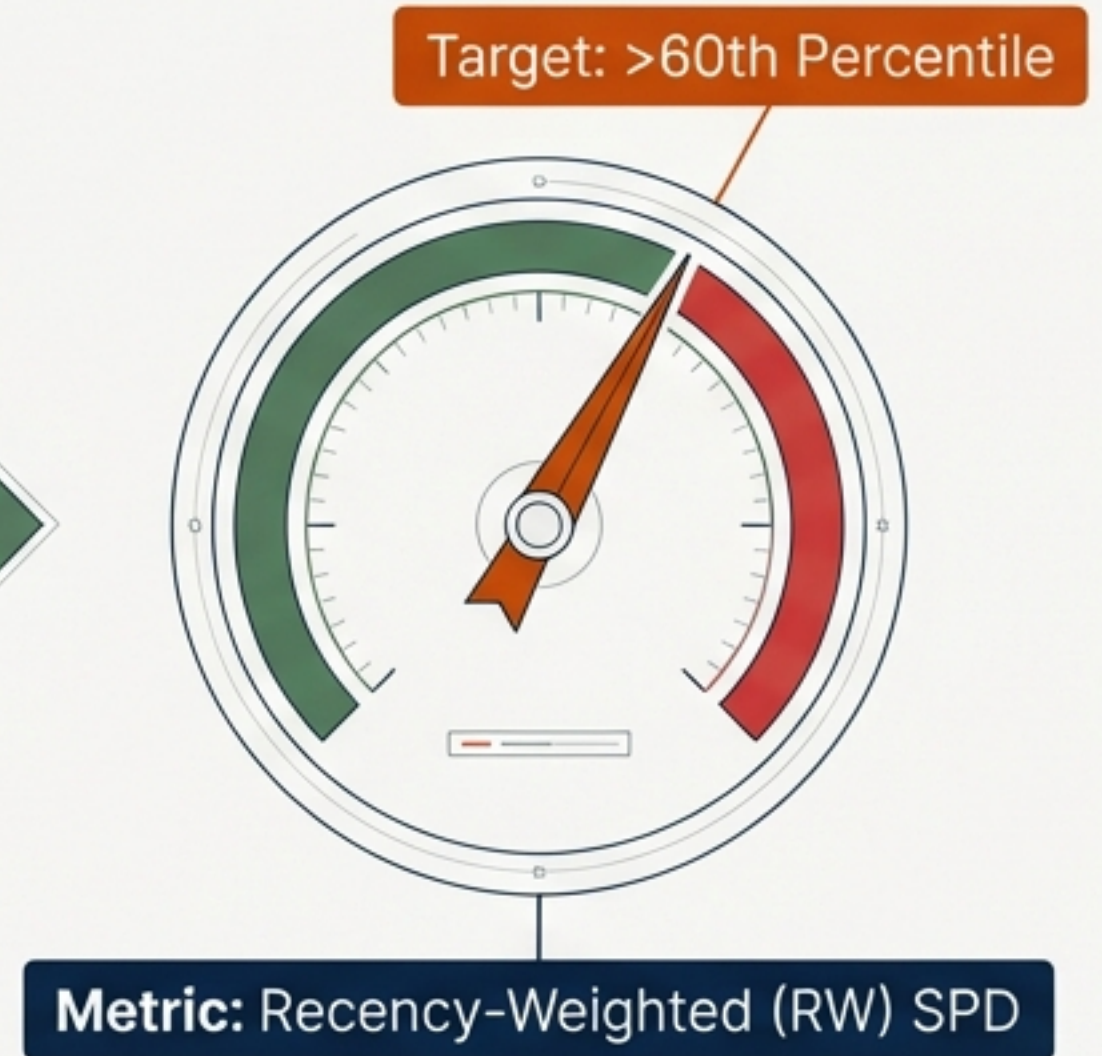
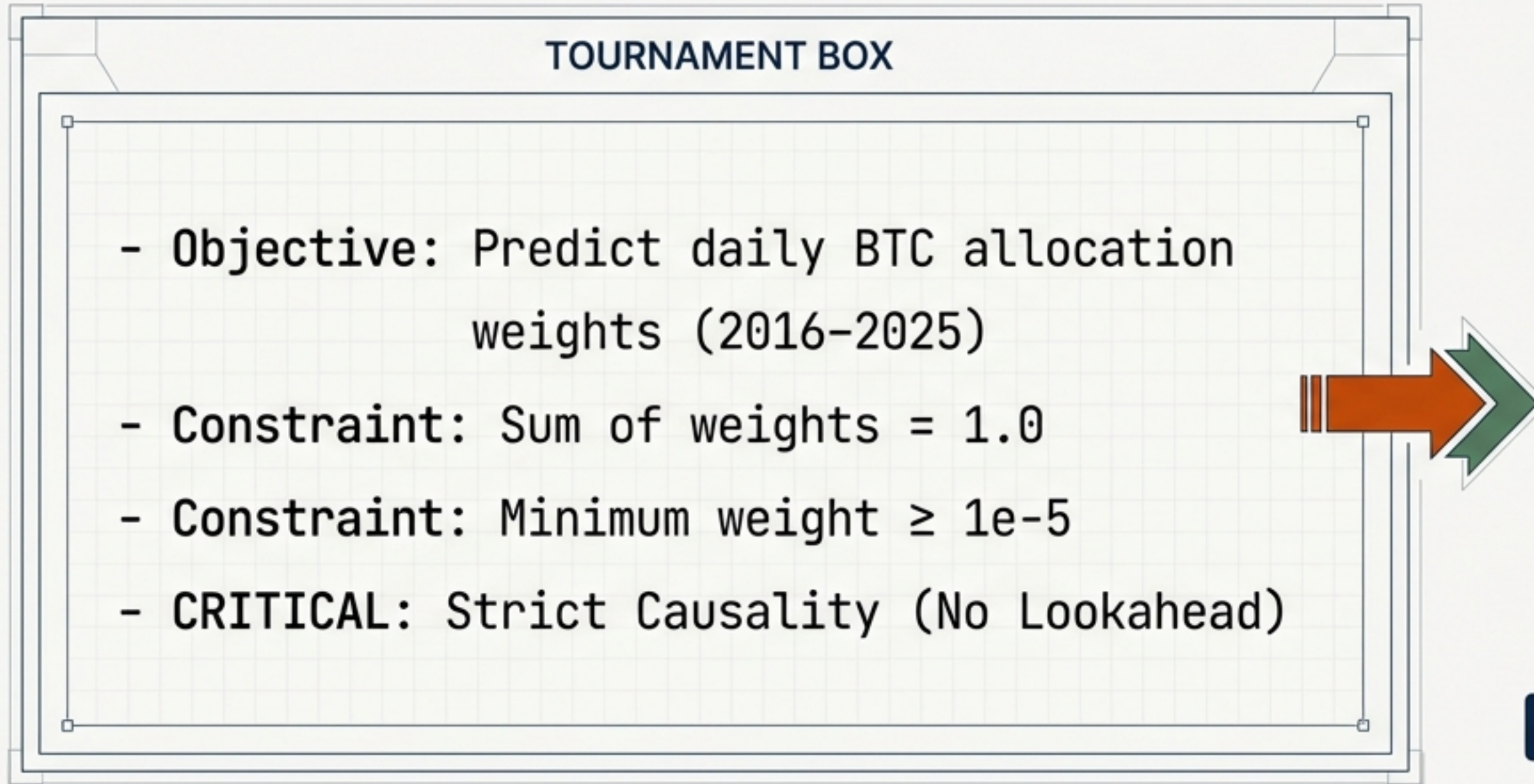


The Risk

Pattern recognition $\neq$
Profitable trading. Can
academic success translate to
strictly causal decisions?

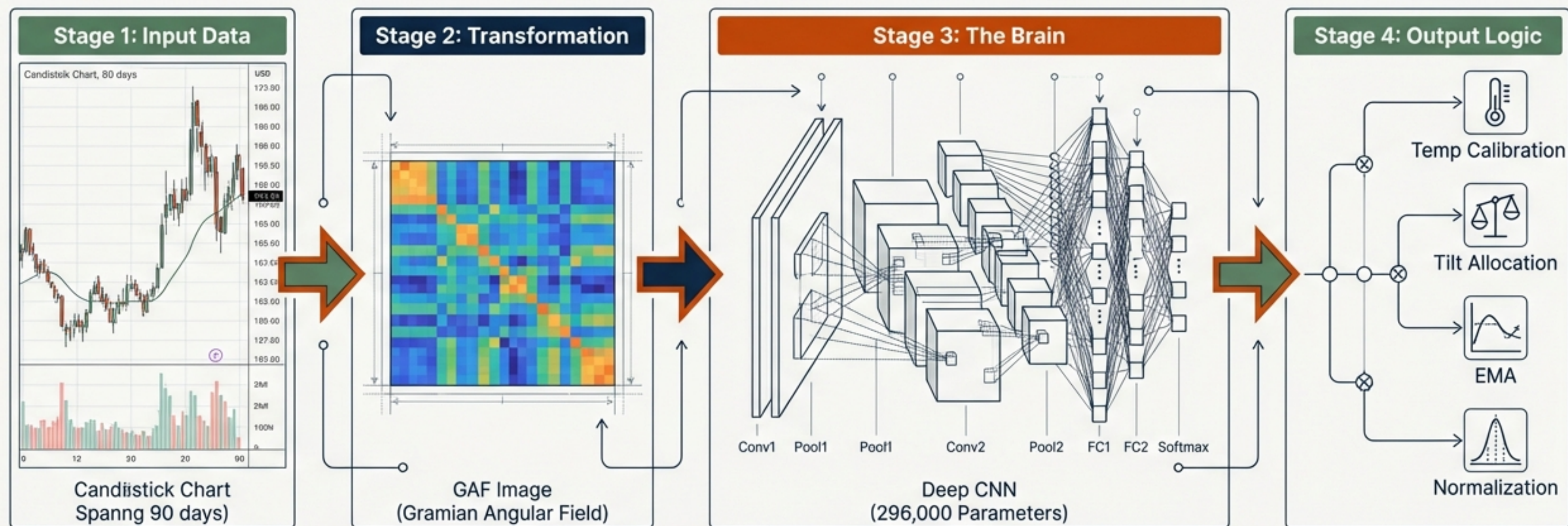
JetBrains Mono
The Reality Gap.

The Arena: VTS Stacking Sats Challenge

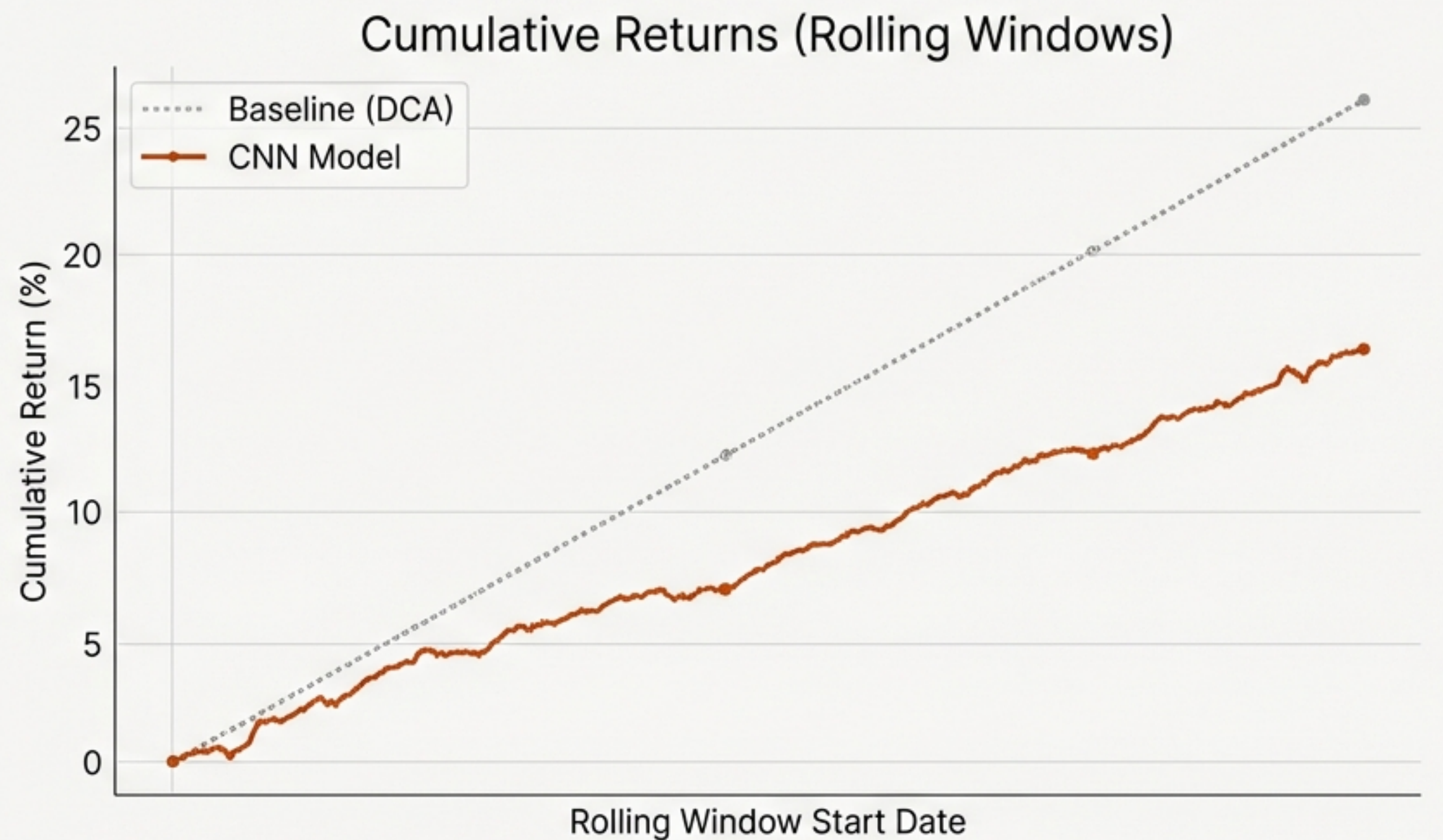


The 296K-Parameter Solution

1.1 MB Model Size | Deep CNN Architecture



Reality Check: The Model Stalls



Evaluated over 3,076 rolling windows.

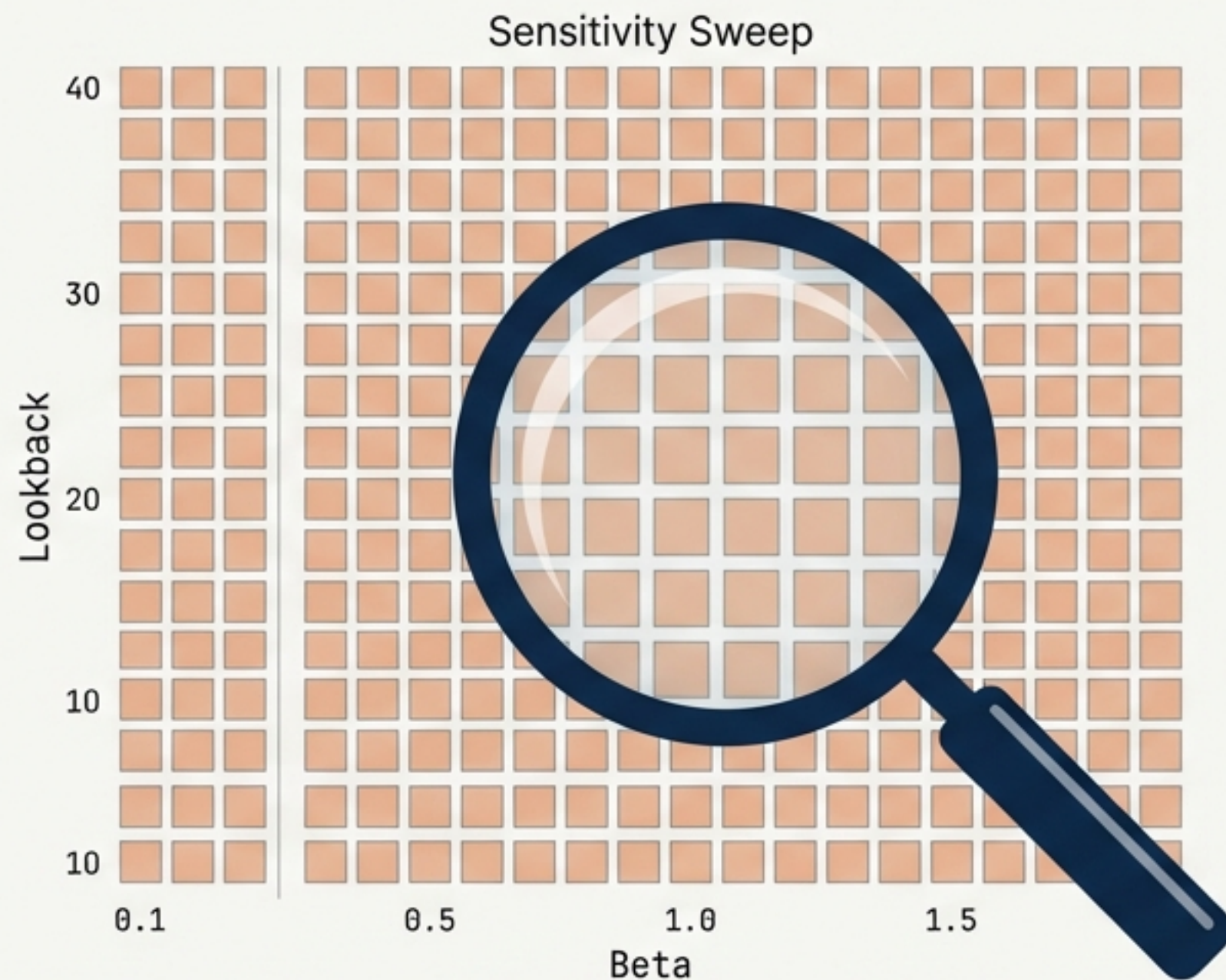
Performance Scorecard

41.43%
RW SPD Percentile
Target >60%

54.32%
Win Rate vs DCA
Barely above chance

Investigation Phase I: Was it the Sizing?

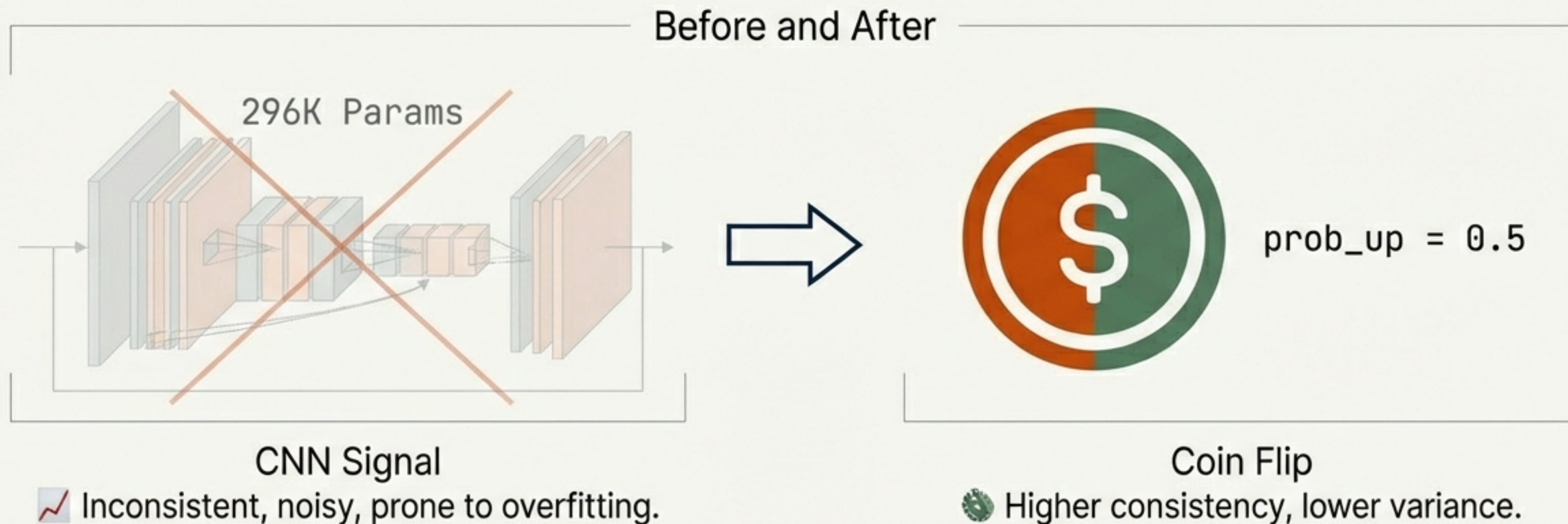
Suspect: Aggressive Allocation Parameters



- **Hypothesis:** Are losses coming from overbetting?
- **Test:** Parameter sensitivity sweep.
- **Result:** Minimal impact. Max improvement +0.35pp.
- **Verdict: Innocent.** The allocator is not the culprit.

Investigation Phase II: The 'Coin Flip' Test

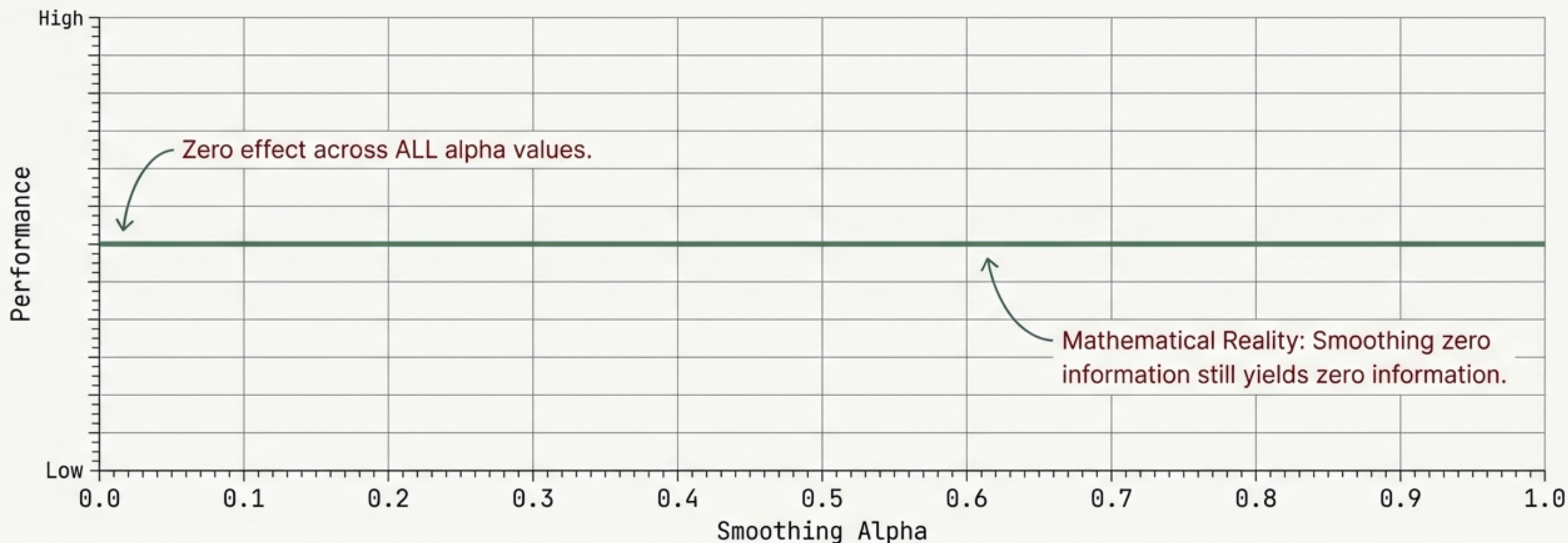
Suspect: The CNN Signal Itself



VERDICT: The CNN signal is effectively worthless.
The complex model was adding noise, not alpha.

Investigation Phase III: The Smoothing Fallacy

Suspect: The Promise of Post-Processing

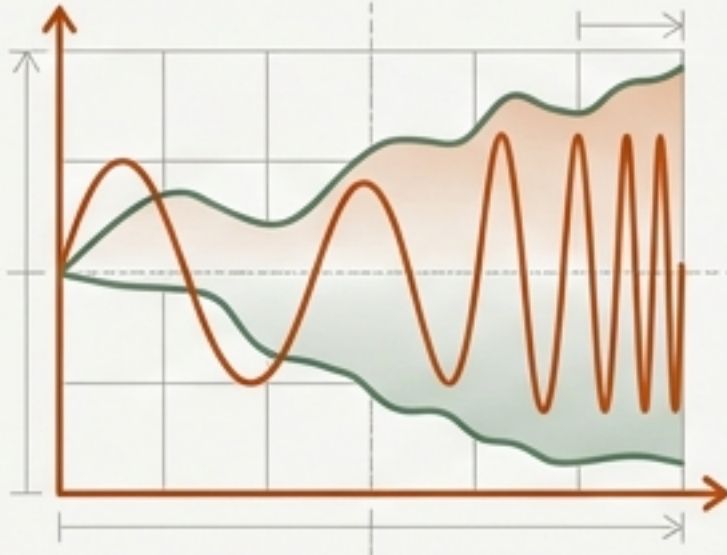


Signal quality is the fundamental issue. No amount of post-processing can fix a bad input.

Diagnosis: Why Visual AI Failed Here

Suspect: The Fundamental Nature of Financial Data

Stationarity



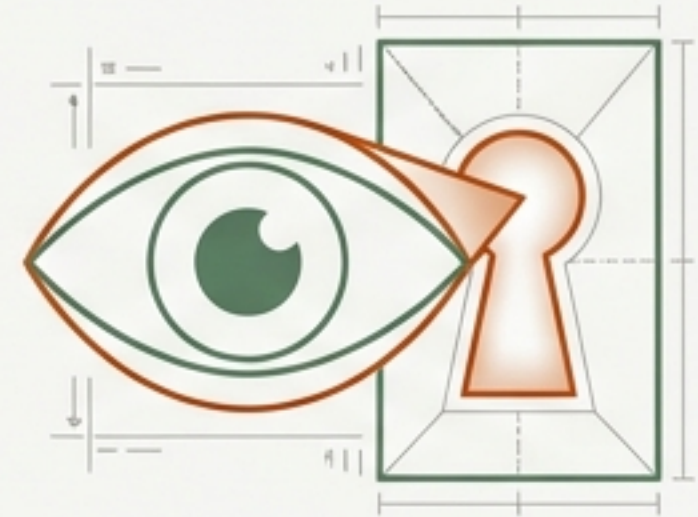
Financial time series change properties over time. Images fail to capture this drift.

Noise vs. Signal



The CNN learned to overfit noise rather than identify predictive structural patterns.

The Lookahead Trap



Academic papers often ignore the strict causality required in live trading tournaments.

The Pivot: A Simplified Baseline

```
def simple_strategy(data):  
    # Neutral probability  
    prob_up = 0.5  
    # Strict causality  
    return allocate(prob_up)
```

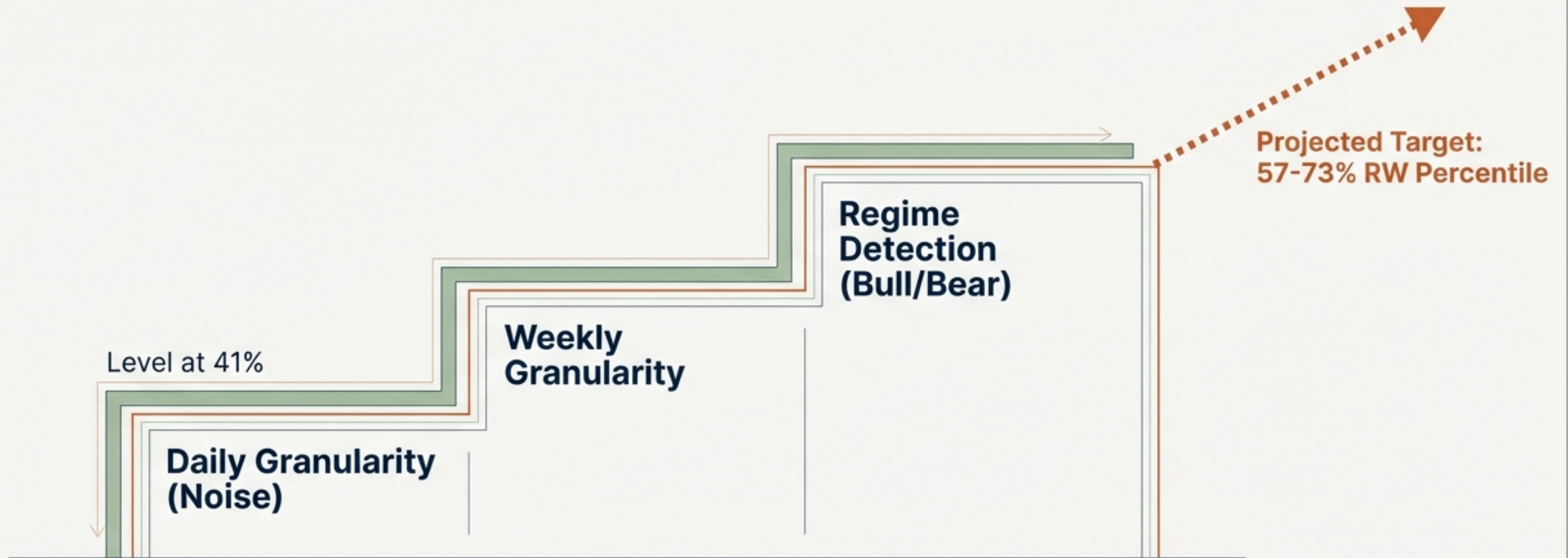
- Dropping the 1.1 MB model entirely
 - Zero model artifacts
 - 10x faster execution
- Performance: 41.94% RW SPD (Higher stability)

Complexity vs. Simplicity

CNN Approach	Simplified Baseline ★
RW SPD: 41.43%	RW SPD: 41.94%
Win Rate: 54.32%	Win Rate: 70.42%
Model Size: 1.1 MB	Model Size: 0 MB
Complexity: HIGH	Complexity: ZERO

Key Insight: The Win Rate jump (54% -> 70%) proves that removing the noisy model allows natural market drift to be handled effectively.

Roadmap to ROI



Moving away from daily noise to identify broader market states.

Key Lessons Learned

1 Ablation is Critical.

Without systematically removing components, we never would have known the CNN was the problem.

2 Test 'Better Than Random' Early.

The coin flip test should be Step 0, not Step 2.

3 Occam's Razor.

296K parameters lost to a coin flip. Complexity is a cost, not a virtue.

4 Scientific Rigor.

Negative results, when properly analyzed, prevent future losses.

Conclusion & Resources

We built a technically sophisticated system and discovered through rigorous testing that it performs equivalently to a coin flip. This simplified version represents our evidence-based conclusion.



Visual_Trading_System
Repo



LESSONS_LEARNED.md



btc_accumulation_model
_simplified.ipynb



“The best model is the one that works, not the one that sounds impressive.”